The Alpha Factory

A miniature strategy factory and a validation gate

Clarion Capital take-home · quantitative trader

Pre-registered criteria `config/gate_criteria.yaml` · SHA-256 `8e35cb92eb5f...d5d8eab` (thresholds unchanged across all strategy iterations; one documented L2 redesign)

Verdict, 48 candidates in, 0 survivors out. The factory builds 48 instances from two opposed root ideas, a volatility-targeted trend follower and a filtered mean-reversion, across four markets. Run as four *independent* layers on the full population, the gate clears no instance through all four. The strongest candidate (the vol-targeted trend on ETH, one of only three instances to clear both the out-of-sample and walk-forward layers, full-sample Sharpe 1.17, individually significant at permutation $p = 0.001$) still fails: it breaches the drawdown cap in the COVID crash, and its **Deflated Sharpe Ratio is** 0.35, far below the 0.95 gate. *Nothing here is alpha.*

1 Two opposed root ideas

The factory ships two families that bet on *opposite* market hypotheses, so a population that "works" cannot be one bet wearing two hats. Each exposes **at most two free parameters**; every other choice is fixed by design, so the optimiser has almost no surface to overfit.

- `voltarget_mom`**, vol-targeted momentum (trend).** The sign of the `lookback`-bar return, with the position scaled by (trailing typical volatility / current `vol_win` volatility): larger in calm, smaller in turbulence. Volatility targeting is the most replicated way to improve a trend signal's risk-adjusted return. The normalising horizon and a leverage cap are fixed; the position is a continuous risk-scaled exposure, which the backtester sizes natively. *Free:* `lookback`, `vol_win`.
- `bollinger_revert`**, filtered mean reversion (reversal).** Fade a deviation of k standard deviations from the `n`-bar mean, *take profit when price crosses back through that mean*, and stay flat between trades. A fixed longer-horizon z-score gates entries, we do *not* fade while a strong trend is underway, the classic way naive reversion dies. *Free:* `n`, `k`.

One bets that moves continue; the other bets that stretched moves revert. Both include an explicit exit and a regime control, so neither is a toy. Architecturally a family is one subclass with a `signal()` method and a parameter grid, registered through `families.register()`; **adding a third root idea is a single `register()` call** and the factory, backtester, gate, figures and report pick it up with no other change, the extensibility the brief asks for.

Which two parameters are free is deliberate. For the trend, `lookback` (the momentum horizon) and `vol_win` (how responsive the volatility estimate is) are freed because neither has a canonical correct value and both directly express the hypothesis; the normalising volatility horizon (a ~3-month baseline, a reference scale) and the leverage cap (a risk limit) are fixed, since optimising a baseline, or letting the backtest raise leverage to chase return, is a textbook overfit. The reversion family is symmetric: `n` (the timescale of a stretched move) and `k` (how many sigmas to fade) are the genuine knobs, while the trend-strength gate (a regime filter) and the mean-cross exit (the definitional close of the trade) are fixed by design rather than tuned.

Population. Each family has a 6-point grid, expanded across four markets, the S&P 500 index (SPXUSD), USD/JPY, gold (XAUUSD) and ETH/USD. That is $2 \times 6 \times 4 = \mathbf{48}$ instances.

2 Realism: data, costs, no look-ahead

Data. The four supplied MetaTrader M1 CSVs (Jan 2020 – mid 2026, ~2.4M rows each) are resampled to **H1**. Hours with no ticks are *dropped, not forward-filled*, preserving genuine gap risk. After cleaning each series carries 38k–40k bars (~6,100 bars/year).

No look-ahead. Signals use only information up to each bar's close, and the backtester applies a strict **one-bar execution lag**: the position over bar t is the signal from bar $t-1$. An explicit test confirms that truncating future data leaves every past signal bit-identical for both families.

Costs. The pre-registered model is the brief's default: the realised half-spread from `<SPREAD>` on each

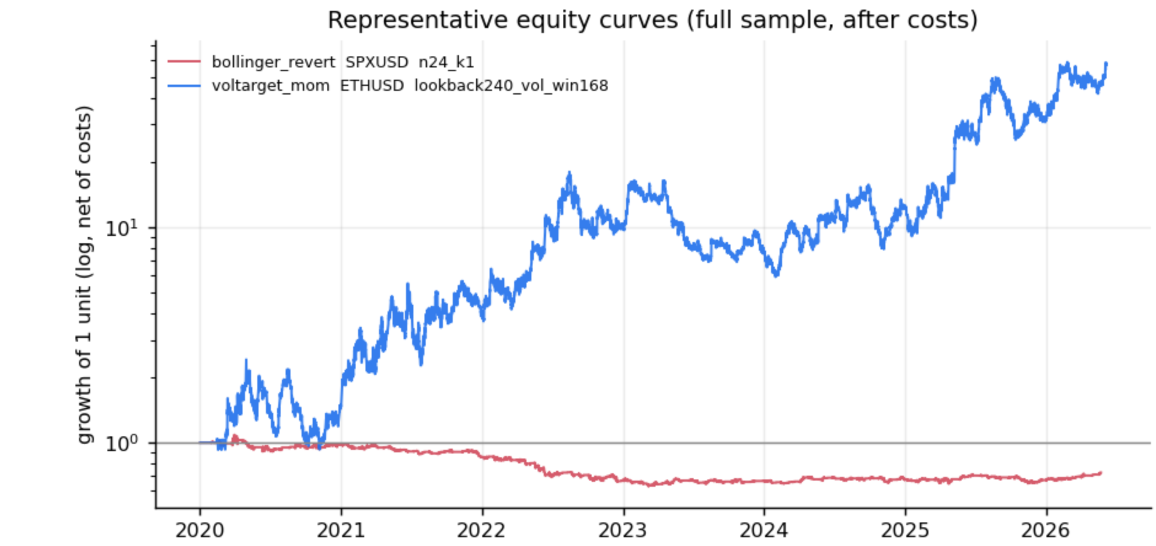

Figure 1: The best instance of each family by out-of-sample Sharpe, net of costs, on a log scale. The vol-targeted trend (blue) is the strongest the factory produced; the best reversion instance (red) barely breaks even net of costs. Both are rejected by the full gate.

side of a position change, plus 0.5 bps/side commission on turnover. Measured mean spreads track the asset hierarchy: USD/JPY 0.57 bps, SPXUSD 0.69 bps, XAUUSD 1.11 bps, ETHUSD 8.18 bps. Costs are what most candidates die against, and the two families fail for opposite reasons (Figure 2). The reversion grid is in fact the *lower*-turnover family (it stays flat between trades) and loses even *gross*: its problem is a weak signal, with costs only a secondary nail. The vol-targeted trend is the *higher*-turnover family (it re-scales every bar) and is where costs bite hardest, eroding a genuine gross edge, on USD/JPY and gold turning a positive gross Sharpe into break-even or loss.

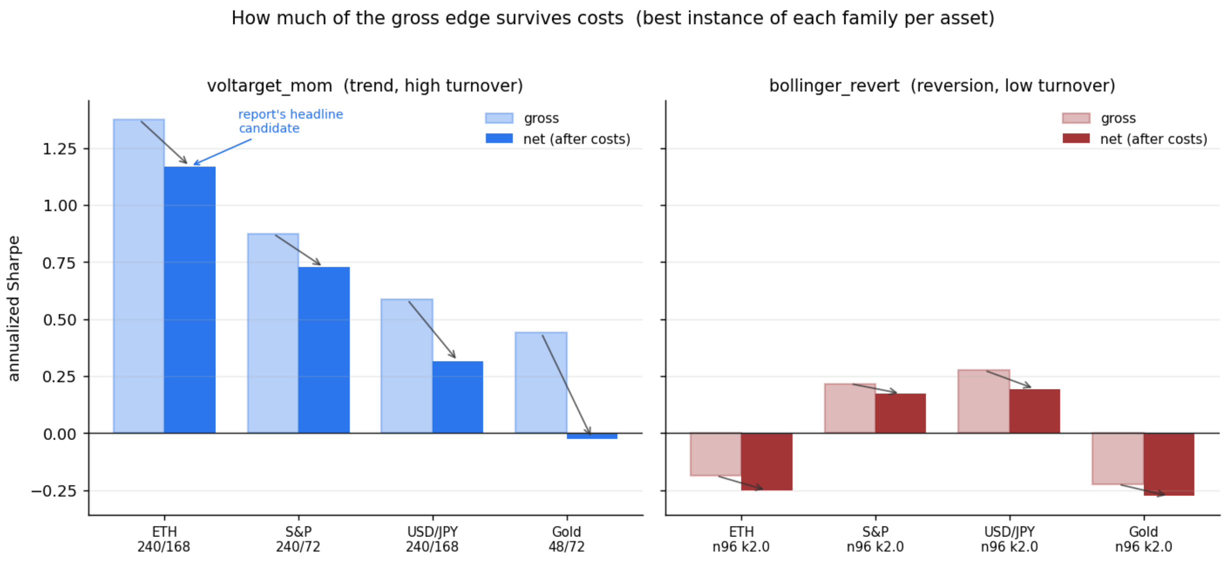

Figure 2: How much of the gross edge survives trading costs: gross vs. net annualised Sharpe for the best instance of each family on each asset (parameters labelled). The trend family (left) has a real gross edge that its high turnover erodes, wiping it out on USD/JPY and gold; the reversion family (right) turns over little, so costs barely bite, but its signal is weak even gross.

3 Pre-registered pass/fail criteria and honesty

Every threshold is fixed in `config/gate_criteria.yaml` *before* the gate runs; `run.py` prints its SHA-256, which is **identical across every strategy iteration** of this project. The strategies changed; the gate never did. The full timeline is in `DECISIONS.md`.

Layer	Test	Pre-registered pass condition
L1	IS/OoS + robustness	OoS Sharpe ≥ 0.50, OoS return > 0, and every $\pm 20\%$ neighbour Sharpe ≥ 0 (median ≥ 0.30)
L2	Walk-forward (rolling windows)	per instance (fixed params): aggregated out-of-window Sharpe ≥ 0.50 and positive in $\geq 60\%$ of windows
L3	Stress replay	over COVID + the instance's own 2 worst 21-day windows: return $\geq -10\%$, DD $\leq 20\%$; Sharpe ≥ 0 under vol-only $2\times$
L4	Monte Carlo + DSR	block-permutation $p < 0.05$ and DSR ≥ 0.95 (N=48)

Each layer is run **independently on all 48 instances**; the funnel is the *intersection* of the four verdicts, not a sequential filter that hides what the later layers would reject.

4 The four-layer gate, what each layer found

4.1 L1, in-sample / out-of-sample + parameter robustness

First 70% in-sample, last 30% out-of-sample; an instance passes only if its OoS performance clears the bar *and* a $\pm 20\%$ perturbation of each parameter leaves no losing neighbour. Because the parameters are *fixed*, the in-sample slice trains nothing, so this is a soft recent-holdout rather than a fit-then-validate step; L1's real anti-overfitting teeth is the $\pm 20\%$ robustness, while the main overfitting source (selecting among 48 designer-chosen configs) is paid for by L2 and, above all, the Deflated Sharpe. **Result: 4/48 pass** (Figure 3), all vol-targeted trends on ETH (OoS Sharpe up to 1.52). The reversion family clears none: faded short-horizon deviations do not survive out-of-sample net of costs. The full parameter landscape (Figure 4) shows the strong region is a broad *plateau* rather than a lone spike, which is exactly what the $\pm 20\%$ robustness check formalises.

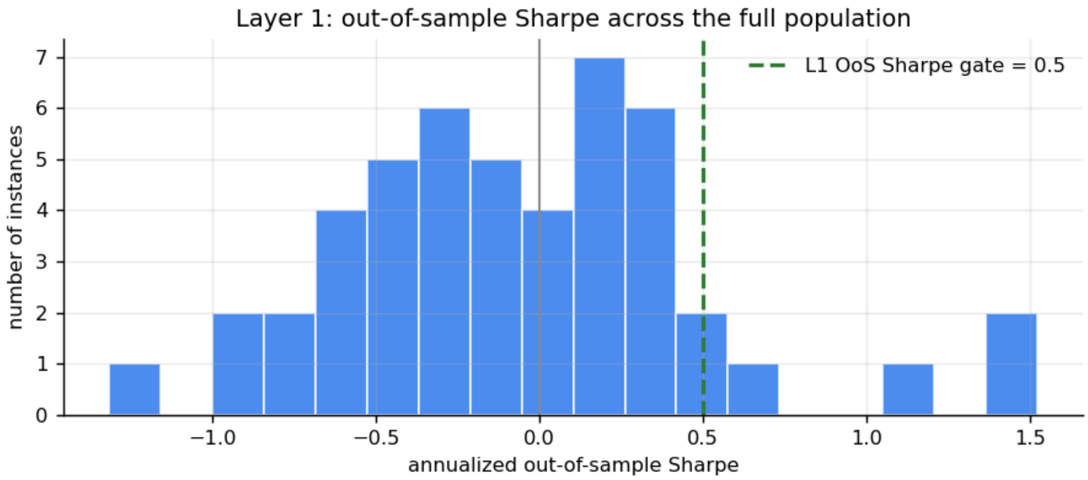

Figure 3: Layer 1: distribution of annualised out-of-sample Sharpe across all 48 instances. Only a thin right tail (vol-targeted trends on ETH) clears the 0.50 gate; the bulk sit at or below zero out-of-sample.

4.2 L2, walk-forward stability (per instance)

With the instance's *fixed* parameters (so nothing is trained), step a 30-week test window through the series (after an initial 90-week span) and judge the instance on its own out-of-window record: aggregated Sharpe ≥ 0.50 *and* a positive return in $\geq 60\%$ of windows. This verdict is independent per instance, the temporal-stability complement to L1's single split (a strategy can pass one and fail the other). **Result: 6/48 pass**, all vol-targeted trends with the long (240-bar) horizon, on ETH and the S&P. A rolling *re-optimisation* (one curve per family/market, plus each instance's parameter-selection frequency) is computed as a *diagnostic* only: an earlier design that gated on the selection frequency was relative across the grid, its pass count was capped by grid arithmetic and a strong stand-alone instance could fail for being the runner-up, so it was demoted (see `DECISIONS.md`).

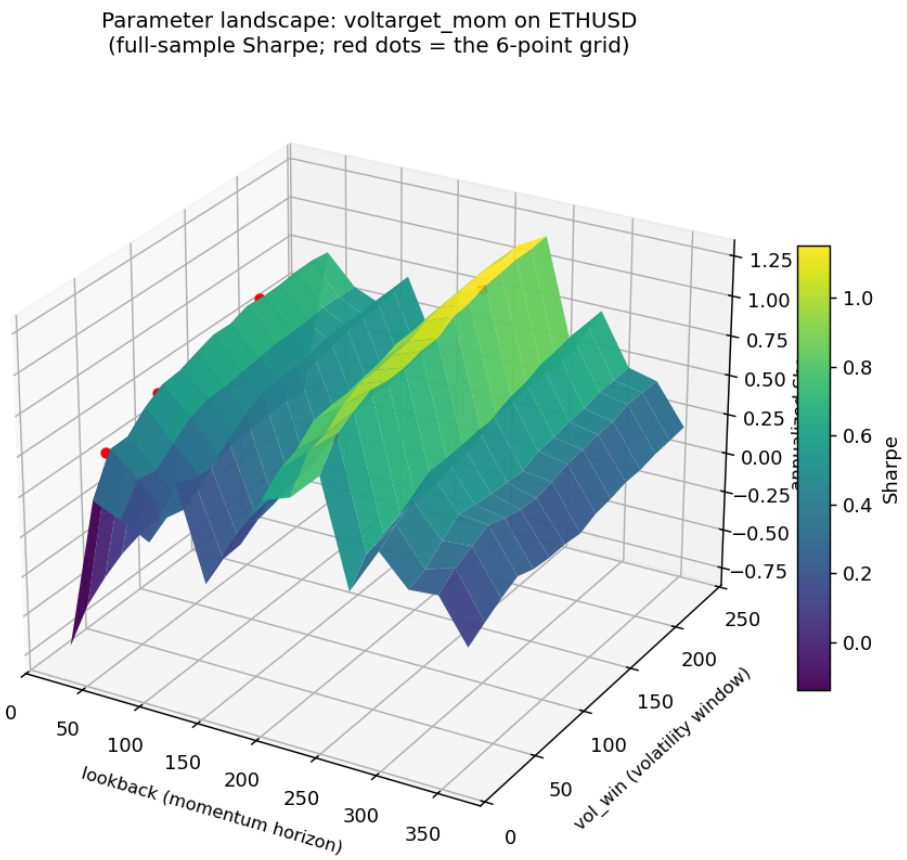

Figure 4: The parameter landscape behind L1's robustness check: full-sample Sharpe of `voltarget_mom` on ETH over a dense `lookback`×`vol_win` mesh (red dots mark the six pre-registered grid points). This is a *diagnostic*, a dense scan visualised here, adding nothing to the $N = 48$ trial count. The profitable region is a broad plateau, not a narrow peak: a Sharpe perched on a spike would be overfit and would fail the ±20% neighbour test.

4.3 L3, stress replay (real + synthetic)

A fixed COVID window plus the two worst rolling 21-day windows *of each instance's own equity*, a momentum book can be short in a crash and profit, so its true worst case is a sharp reversal a market-drawdown window would miss, and a *volatility-only* 2× synthetic path (only the deviations are scaled, the drift held, so it does not flatter trend). "Acceptable" (return $\geq -10\%$, DD $\leq 20\%$, synthetic Sharpe ≥ 0) was fixed in advance. **Result: 8/48 pass**, all on USD/JPY, the calmest market. The dominant kill is the COVID window; the own-worst-window test also exposes a 47% drawdown in the ETH trend's worst 21 days, and L3 removes the three instances that had cleared both L1 and L2.

4.4 L4, Monte Carlo + Deflated Sharpe Ratio

A *block*-permutation test against a no-skill null, shuffling the order of ∼1-day return blocks, so the null keeps the volatility clustering an i.i.d. shuffle would erase, a moving-block bootstrap (block = 24 bars) for the drawdown distribution (a *diagnostic*, not gated), and the Deflated Sharpe Ratio with `n_trials` set to the *total number of instances*, $N = 48$. The permutation only asks whether the edge is real on *this* series; paying for having tried 48 of them is the Deflated Sharpe's job. **Result: 0/48 pass**, the subject of §6.

5 Survival funnel

Per-layer view (each layer judged independently on all 48)

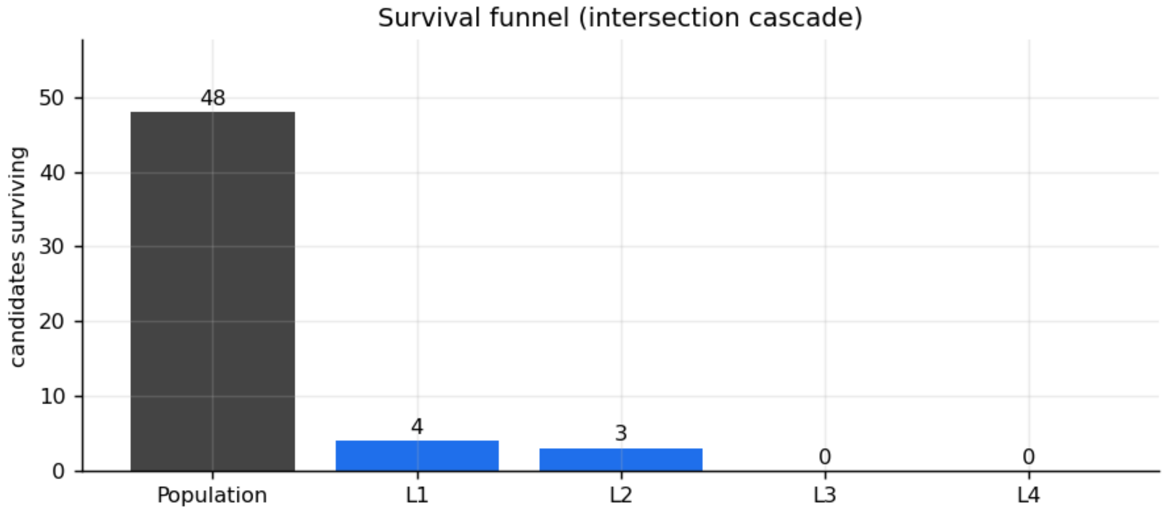

Figure 5: Intersection cascade. L1 admits 4 instances; three of them also clear the walk-forward stability test (L2), and the COVID stress window (L3) then removes all three. The intersection of all four layers is empty.

Layer	Test	Pass	Fail	Dominant failure reason
L1	IS/OoS + robustness	4	44	OoS Sharpe below threshold
L2	Walk-forward	6	42	out-of-window Sharpe/return below bar
L3	Stress replay	8	40	blows up in COVID window
L4	Monte Carlo + DSR	0	48	no skill (permutation) *and* fails DSR
Intersection (all four)		**0**	**48**	—

Three instances clear both L1 and L2 (the long-horizon ETH vol-targeted trends), and no instance clears three layers: a holdout split and a rolling-stability test agree only on a handful of trend instances, and none of those survives the COVID crisis.

6 The Deflated Sharpe verdict

This is the heart of the exercise. We did not test one strategy; we tested 48. The expected maximum Sharpe of $N = 48$ zero-skill trials is, on this data, a per-observation $\mathrm{SR}^\star \approx 0.0168$, an **annualised hurdle near** 1.3 that a strategy must clear before its Sharpe is even evidence of skill.

The sharpest illustration is the best candidate itself. The ETH vol-targeted trend is one of three instances to pass both L1 and L2, with a full-sample Sharpe of 1.17, a permutation $p = 0.001$, and positive returns in most bootstrap resamples. Judged alone it looks real. Judged honestly, as one of 48 things we tried, and against the COVID crash, it is rejected twice over: its Deflated Sharpe is 0.35, and it breaches the 20% drawdown cap in 2020. The multiple-testing correction is not a footnote; it is the verdict, and it is why "just generate more candidates" cannot manufacture a survivor, every instance added only raises the bar (`n_trials`) for all of them. The flip side is that the Deflated Sharpe depends on the *declared* trial count: a Sharpe earned by one pre-committed hypothesis is genuinely stronger evidence than the same number cherry-picked as the best of 48, so the only way to flatter it is to *under-report* N, which is dishonesty rather than method and is auditable here from the code that builds all 48. The principled adjustment runs the other way, through the effective trial count below.

7 Verdict, limitations, and honesty

Verdict: deploy nothing. On this data with these (pre-registered, demanding) criteria, $48 \to 0$ is the right answer. The factory manufactured plausible candidates, including one genuinely convincing in isolation, and the gate refused to be impressed by any of them.

What would change the verdict. A deployable instance would have to clear OoS *and* walk-forward (L1∩L2), hold its floor through COVID and a 2×-vol path (L3), and post a Deflated Sharpe ≥ 0.95, which at $N = 48$ demands an annualised, stress-robust Sharpe well above the 1.17 the best candidate

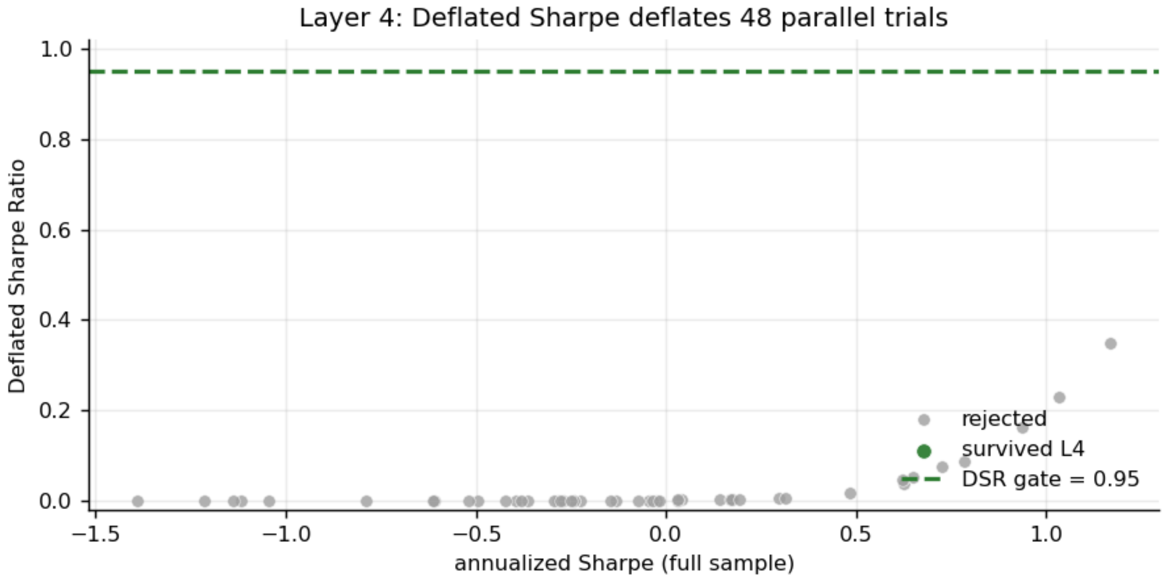

Figure 6: Deflated Sharpe vs. raw annualised Sharpe across all 48 instances. The best DSR is 0.35 (the ETH vol-targeted trend, raw Sharpe 1.17); the 0.95 gate is never reached. Higher raw Sharpe buys very little once 48 parallel trials are deflated.

reached.

Honesty. The strategies were iterated during development (a dual-SMA/z-score pair, then a multi-timeframe momentum/breakout pair, a vol-targeting family added to demonstrate extensibility, and finally the two opposed families here); across all four iterations the `gate_criteria.yaml` SHA-256 was *identical*, the proof that no threshold was ever tuned to admit a candidate. The config then changed once, deliberately, when L2's *methodology* was redesigned: its original verdict re-optimised the grid and judged each instance *relative* to its siblings (capped by grid arithmetic, decided by a noisy in-sample ranking), so a strong stand-alone instance could fail merely for being the runner-up. L2 is now an *independent* per-instance stability test (fixed parameters, consistency across rolling out-of-window windows), with the re-optimisation retained as a diagnostic. The change is documented, raised L2's pass count from 1 to 6, and *left the verdict unchanged* ($48 \to 0$), confirming it fixed a design flaw rather than admitting anything. Two further methodology refinements touched no threshold (the SHA-256 is unchanged): L3 now stresses each instance over its *own* worst 21-day windows (a market-drawdown window misses a strategy whose worst case is a sharp reversal) and uses a volatility-only synthetic shock (doubling volatility without flattering trend), and L4's significance test is now a block permutation that preserves serial structure (an i.i.d. shuffle was anti-conservative). Both left the verdict unchanged. `DECISIONS.md` records the full timeline.

Correlated trials and effective N. The Deflated Sharpe uses the raw trial count $N = 48$, but the 48 instances are not independent. Within each family/market the six grid points are highly correlated (mean pairwise return correlation ≈ 0.60), while across the two opposed families the return correlation is essentially nil (≈ -0.09); their *signals* are negatively correlated (≈ -0.30) by construction, which is what makes the two families genuine diversification rather than one bet in disguise. A participation-ratio count of the 48×48 return-correlation matrix, $(\sum_i \lambda_i)^2 / \sum_i \lambda_i^2$, gives an effective $N \approx 12$ (12.1, and stable at 12.1 whether returns are aligned hourly, daily, or on the common index), not 48. Using $N \approx 12$ raises the best instance's Deflated Sharpe from 0.35 to 0.69, still well below the 0.95 gate. We report the conservative raw $N = 48$; the verdict is unchanged under either count, and an effective-N deflation is the natural refinement for a future iteration. The figure is reproduced by `effective_n.py`.

Limitations. The simplifications are deliberate: two families, coarse grids, one timeframe (H1), a mean-spread cost model, and the conservative raw N. The path forward is not a richer factory (more variations only raise the deflation bar) but more history, genuinely orthogonal alpha, and portfolio-level combination of weak uncorrelated signals, never a looser gate.